§6 Hyperbolic Functions

■ Hyperbolic Functions and Their Derivatives

*** Definition of the Hyperbolic Functions**

(1) $\sinh x = \dfrac{e^x - e^{-x}}{2}$: hyperbolic sine function

(2) $\cosh x = \dfrac{e^x + e^{-x}}{2}$: hyperbolic cosine function

(3) $\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$: hyperbolic tangent function

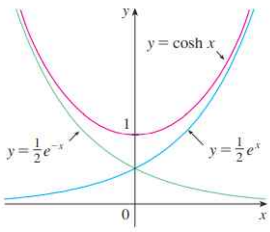
(4) $\coth x = \dfrac{1}{\tanh x} = \dfrac{\cosh x}{\sinh x} = \dfrac{e^x + e^{-x}}{e^x - e^{-x}}$: hyperbolic cotangent function

(5) $\operatorname{sech} x = \dfrac{1}{\cosh x} = \dfrac{2}{e^x + e^{-x}}$: hyperbolic secant function

(6) $\operatorname{csch} x = \dfrac{1}{\sinh x} = \dfrac{2}{e^x - e^{-x}}$: hyperbolic cosecant function

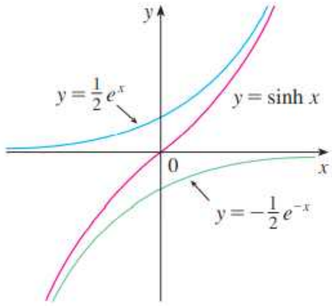
*** Graph of the Hyperbolic Functions**

(1) $y = \cosh x = \dfrac{e^x + e^{-x}}{2}$



- Range: $\{y \mid y \geq 1\}$
- Even function

(2) $y = \sinh x = \dfrac{e^x - e^{-x}}{2}$

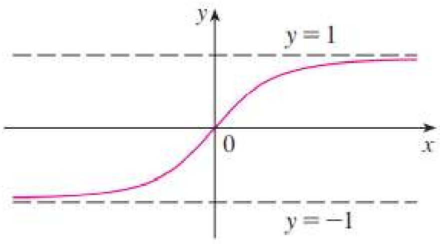


- Range: $\mathbb{R} = \{y \mid -\infty < y < \infty\}$
- Odd function

(3) $y = \tanh x = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

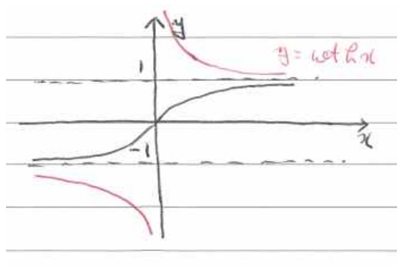
- $|e^x - e^{-x}| < |e^x + e^{-x}| \Rightarrow \left|\dfrac{e^x - e^{-x}}{e^x + e^{-x}}\right| < 1 \Rightarrow -1 < \tanh x < 1$
- $\displaystyle\lim_{x\to\infty} \tanh x = \lim_{x\to\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = \lim_{x\to\infty} \frac{1 - e^{-2x}}{1 + e^{-2x}} = 1$

  $\displaystyle\lim_{x\to-\infty} \tanh x = \lim_{x\to-\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = \lim_{x\to-\infty} \frac{e^{2x} - 1}{e^{2x} + 1} = -1$



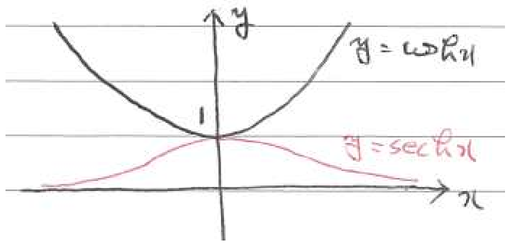
- Range: $\{y \mid -1 < y < 1\}$
- Odd function
- Asymptotes: $y = \pm 1$

(4) $y = \coth x = \dfrac{1}{\tanh x}$



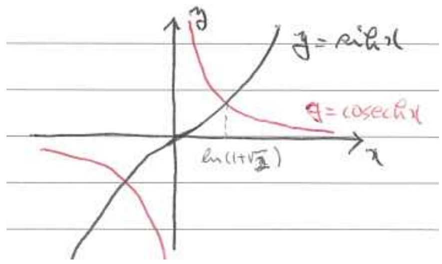
- Domain: $\{x \in \mathbb{R} \mid x \neq 0\}$
- Range: $\{y \mid y < -1,\quad y > 1\}$
- Odd function
- Asymptotes: $y = \pm 1$, $x = 0$

(5) $y = \operatorname{sech} x = \dfrac{1}{\cosh x}$



- Range: $\{y \mid 0 < y \leq 1\}$
- Even function
- Asymptotes: $y = 0$

(6) $y = \operatorname{csch} x = \dfrac{1}{\sinh x}$



- Domain: $\{x \in \mathbb{R} \mid x \neq 0\}$
- Range: $\{y \in \mathbb{R} \mid y \neq 0\}$
- Odd function
- Asymptotes: $y = 0$, $x = 0$

* Properties of the Hyperbolic Functions

(1) $\sinh(-x) = -\sinh x$

$\cosh(-x) = \cosh x$

$\tanh(-x) = -\tanh x$

$\coth(-x) = -\coth x$

$\operatorname{sech}(-x) = \operatorname{sech} x$

$\operatorname{csch}(-x) = -\operatorname{csch} x$

(2) $\cosh^2 x - \sinh^2 x = 1$

$1 - \tanh^2 x = \operatorname{sech}^2 x$

$\coth^2 x - 1 = \operatorname{csch}^2 x$

(pf)

(3) $\sinh(x \pm y) = \sinh x \cosh y \pm \sinh y \cosh x$

$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$

$$\tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$$

(pf)

(4) $\sinh(2x) = 2\sinh x \cosh x$

$$\begin{aligned}\cosh(2x) &= \cosh^2 x + \sinh^2 x \\ &= 2\cosh^2 x - 1 \\ &= 1 + 2\sinh^2 x\end{aligned}$$

$$\tanh(2x) = \frac{2\tanh x}{1 + \tanh^2 x}$$

(5) $\cosh^2 x = \dfrac{\cosh(2x) + 1}{2}$

$$\sinh^2 x = \frac{\cosh(2x) - 1}{2}$$

$$\tanh^2 x = \frac{\cosh(2x) - 1}{\cosh(2x) + 1}$$

* Derivatives of the Hyperbolic Functions

(1) $y = \sinh x = \dfrac{e^x - e^{-x}}{2}$

$\Rightarrow \; y' = \dfrac{e^x + e^{-x}}{2} = \cosh x$

(2) $y = \cosh x = \dfrac{e^x + e^{-x}}{2}$

$\Rightarrow \; y' = \dfrac{e^x - e^{-x}}{2} = \sinh x$

(3) $y = \tanh x = \dfrac{\sinh x}{\cosh x}$

$$
\begin{aligned}
\Rightarrow \; y' &= \frac{(\sinh x)' \cosh x - \sinh x (\cosh x)'}{\cosh^2 x} \\
&= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} \\
&= \frac{1}{\cosh^2 x} \\
&= \operatorname{sech}^2 x
\end{aligned}
$$

(4) $y = \coth x = \dfrac{\cosh x}{\sinh x}$

$$
\begin{aligned}
\Rightarrow \; y' &= \frac{(\cosh x)' \sinh x - \cosh x (\sinh x)'}{\sinh^2 x} \\
&= \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x} \\
&= \frac{-1}{\sinh^2 x} \\
&= -\operatorname{csch}^2 x
\end{aligned}
$$

(5) $y = \mathrm{sech}x = \dfrac{1}{\cosh x} = (\cosh x)^{-1}$

$$\begin{aligned}
\Rightarrow\ y' &= -(\cosh x)^{-2}(\cosh x)' \\
&= -\frac{1}{\cosh^2 x}\sinh x \\
&= -\frac{1}{\cosh x}\,\frac{\sinh x}{\cosh x} \\
&= -\mathrm{sech}x\,\tanh x
\end{aligned}$$

(6) $y = \mathrm{csch}x = \dfrac{1}{\sinh x} = (\sinh x)^{-1}$

$$\begin{aligned}
\Rightarrow\ y' &= -(\sinh x)^{-2}(\sinh x)' \\
&= -\frac{1}{\sinh^2 x}\cosh x \\
&= -\frac{1}{\sinh x}\,\frac{\cosh x}{\sinh x} \\
&= -\mathrm{csch}x\,\coth x
\end{aligned}$$

Exam 2 If $y = \cosh\sqrt{x}$, find $\dfrac{dy}{dx}$.

(sol)

■ Inverse Hyperbolic Functions and Their Derivatives

＊ Definition of Inverse Hyperbolic Functions

(1) $y = \sinh x$

$\xrightarrow{inverse}$ $x = \sinh y$

$\Leftrightarrow$ $y = \sinh^{-1} x$: inverse hyperbolic sine function

(2) $y = \cosh x \quad (x \geq 0)$

$\xrightarrow{inverse}$ $x = \cosh y \quad (y \geq 0)$

$\Leftrightarrow$ $y = \cosh^{-1} x$: inverse hyperbolic cosine function

(3) $y = \tanh x$

$\xrightarrow{inverse}$ $x = \tanh y$

$\Leftrightarrow$ $y = \tanh^{-1} x$: inverse hyperbolic tangent function

(4) $y = \coth x$

$\xrightarrow{inverse}$ $x = \coth y$

$\Leftrightarrow$ $y = \coth^{-1} x$: inverse hyperbolic cotangent function

(5) $y = \operatorname{sech} x \quad (x \geq 0)$

$\xrightarrow{inverse}$ $x = \operatorname{sech} y \quad (y \geq 0)$

$\Leftrightarrow$ $y = \operatorname{sech}^{-1} x$: inverse hyperbolic secant function

(6) $y = \operatorname{csch} x$

$\xrightarrow{inverse}$ $x = \operatorname{csch} y$

$\Leftrightarrow$ $y = \operatorname{csch}^{-1} x$: inverse hyperbolic cosecant function

∗ Logarithm Expression of Inverse Hyperbolic Functions

(1) $y = \sinh^{-1} x \quad \Leftrightarrow \quad x = \sinh y = \dfrac{e^y - e^{-y}}{2}$

$$
\begin{aligned}
&\Rightarrow\ e^y - e^{-y} = 2x \\
&\Rightarrow\ (e^y)^2 - 2xe^y - 1 = 0 \\
&\Rightarrow\ e^y = x + \sqrt{x^2+1} \qquad (\because e^y > 0) \\
&\Rightarrow\ y = \ln\left(x + \sqrt{x^2+1}\right)
\end{aligned}
$$

(2) $y = \cosh^{-1} x \quad \Leftrightarrow \quad x = \cosh y = \dfrac{e^y + e^{-y}}{2} \quad (y \geq 0)$

$$
\begin{aligned}
&\Rightarrow\ e^y + e^{-y} = 2x \\
&\Rightarrow\ (e^y)^2 - 2xe^y + 1 = 0 \\
&\Rightarrow\ e^y = x + \sqrt{x^2-1} \qquad (\because\ e^y \geq 1 \quad \text{if} \quad y \geq 0) \\
&\Rightarrow\ y = \ln\left(x + \sqrt{x^2-1}\right)
\end{aligned}
$$

(3) $y = \tanh^{-1} x \quad \Leftrightarrow \quad x = \tanh y = \dfrac{e^y - e^{-y}}{e^y + e^{-y}}$

$$
\begin{aligned}
&\Rightarrow\ x e^y + x e^{-y} = e^y - e^{-y} \\
&\Rightarrow\ e^y(1-x) = (1+x)e^{-y} \\
&\Rightarrow\ e^{2y} = \frac{1+x}{1-x} \\
&\Rightarrow\ 2y = \ln\frac{1+x}{1-x} \\
&\Rightarrow\ y = \frac{1}{2}\ln\frac{1+x}{1-x}
\end{aligned}
$$

(4) $y = \coth^{-1} x \quad \Leftrightarrow \quad x = \coth y = \dfrac{e^y + e^{-y}}{e^y - e^{-y}}$

$$
\begin{aligned}
&\Rightarrow\ x e^y - x e^{-y} = e^y + e^{-y} \\
&\Rightarrow\ e^y(x-1) = (x+1)e^{-y} \\
&\Rightarrow\ e^{2y} = \frac{x+1}{x-1} \\
&\Rightarrow\ 2y = \ln\frac{x+1}{x-1} \\
&\Rightarrow\ y = \frac{1}{2}\ln\frac{x+1}{x-1}
\end{aligned}
$$

(5) $y = \operatorname{sech}^{-1} x \quad \Leftrightarrow \quad x = \operatorname{sech} y = \dfrac{1}{\cosh y} \quad (y \geq 0)$

$$\Rightarrow \cosh y = \frac{1}{x}$$

$$\Rightarrow y = \cosh^{-1}\left(\frac{1}{x}\right)$$

$$= \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1}\right)$$

(6) $y = \operatorname{csch}^{-1} x \quad \Leftrightarrow \quad x = \operatorname{csch} y = \dfrac{1}{\sinh y}$

$$\Rightarrow \sinh y = \frac{1}{x}$$

$$\Rightarrow y = \sinh^{-1}\left(\frac{1}{x}\right)$$

$$= \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right)$$

* Derivatives of the Inverse Hyperbolic Functions

(1) $y=\sinh^{-1}x \quad \Leftrightarrow \quad x=\sinh y$

$$\Rightarrow\ 1=\cosh y\, y'$$
$$\Rightarrow\ y'=\frac{1}{\cosh y}$$
$$=\frac{1}{\sqrt{\sinh^2 y+1}} \quad (\because\ \cosh y\geq 0)$$
$$=\frac{1}{\sqrt{x^2+1}}$$

(2) $y=\cosh^{-1}x \quad \Leftrightarrow \quad x=\cosh y \quad (y\geq 0)$

$$\Rightarrow\ 1=\sinh y\, y'$$
$$\Rightarrow\ y'=\frac{1}{\sinh y}$$
$$=\frac{1}{\sqrt{\cosh^2 y-1}} \quad (\because\ \sinh y\geq 0 \quad \text{if} \quad y\geq 0)$$
$$=\frac{1}{\sqrt{x^2-1}}$$

(3) $y=\tanh^{-1}x \quad \Leftrightarrow \quad x=\tanh y$

$$\Rightarrow\ 1=\operatorname{sech}^2 y\, y'$$
$$\Rightarrow\ y'=\frac{1}{\operatorname{sech}^2 y}$$
$$=\frac{1}{1-\tanh^2 y}$$
$$=\frac{1}{1-x^2}$$

(4) $y=\coth^{-1}x \quad \Leftrightarrow \quad x=\coth y$

$$\Rightarrow\ 1=-\operatorname{csch}^2 y\, y'$$
$$\Rightarrow\ y'=-\frac{1}{\operatorname{csch}^2 y}$$
$$=-\frac{1}{\coth^2 y-1}$$
$$=-\frac{1}{x^2-1}$$

(5) $y = \mathrm{sech}^{-1}x \quad \Leftrightarrow \quad x = \mathrm{sech}\,y = \dfrac{1}{\cosh y} \quad (y \geq 0)$

$$\begin{aligned}
&\Rightarrow\ y = \cosh^{-1}\left(\frac{1}{x}\right) \\
&\Rightarrow\ y' = \frac{1}{\sqrt{\left(\frac{1}{x}\right)^2 - 1}}\left(\frac{1}{x}\right)' \\
&\qquad = \frac{x}{\sqrt{1-x^2}}\left(-\frac{1}{x^2}\right) \\
&\qquad = -\frac{1}{x\sqrt{1-x^2}}
\end{aligned}$$

(6) $y = \mathrm{csch}^{-1}x \quad \Leftrightarrow \quad x = \mathrm{csch}\,y = \dfrac{1}{\sinh y}$

$$\begin{aligned}
&\Rightarrow\ y = \sinh^{-1}\left(\frac{1}{x}\right) \\
&\Rightarrow\ y' = \frac{1}{\sqrt{1+\left(\frac{1}{x}\right)^2}}\left(\frac{1}{x}\right)' \\
&\qquad = \frac{|x|}{\sqrt{x^2+1}}\left(-\frac{1}{x^2}\right) \\
&\qquad = -\frac{1}{|x|\sqrt{x^2+1}}
\end{aligned}$$

Exam 5 Find $\dfrac{d}{dx}[\tanh^{-1}(\sin x)]$.

(sol)

(Exam) Differentiate the following functions

1. $y = x^2 \ln \sinh x^2$

2. $y = \operatorname{sech}^3\left(\frac{3}{2}x\right)$

3. $y = \cosh^{-1}(\sec x), \quad 0 \le x \le \frac{\pi}{2}$

(sol)

(Exam) Find y' if $x \cosh y + y \sinh x = 1$

(sol)

(Exam) Find y' if $\cosh(x+y) = x - y$

(sol)